

Abhilesh Kumar

Raipur, India

+91-6387381984 abhileshk68@gmail.com LinkedIn GitHub

Education

• National Institute of Technology (NIT) Raipur	2022 – 2026
.Tech in Biotechnology Engineering	CGPA:7.04
• National Inter College, Maudaha, Hamirpur (UP Board)	2021
Senior Secondary (Class XII)	74.8%
• National Inter College, Maudaha, Hamirpur (UP Board)	2019
Secondary (Class X)	78.6%

Technical Skills

Programming: Python, SQL

Machine Learning: Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Cross Validation

Deep Learning: Neural Networks, LSTM, Basic Model Training

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib, Seaborn

AI / NLP: Natural Language Processing, Prompt Engineering, Large Language Models (LLM) Basics

Tools: Jupyter Notebook, Git, GitHub, VS Code, Data Visualization

AI / ML Projects

Movie Recommender System — Python, Machine Learning

- Built a content-based movie recommendation system using the TMDB dataset containing 5000+ movies.
- Applied TF-IDF vectorization and cosine similarity to analyze movie metadata including genres, keywords, and cast.
- Designed a recommendation pipeline that suggests top similar movies based on user input.
- Improved recommendation relevance by implementing text preprocessing and feature engineering techniques.

CRISPR Off-Target Prediction — Machine Learning, Python

- Developed a machine learning model to predict CRISPR-Cas9 genome editing off-target sites.
- Used GUIDE-seq and CIRCLE-seq datasets to analyze potential unintended DNA edits.
- Implemented Random Forest classification model and evaluated performance using ROC-AUC and accuracy metrics.
- Performed data preprocessing, feature extraction and model evaluation to improve prediction reliability.

Customer Churn Prediction — Python, Machine Learning

- Built predictive models to identify customers likely to leave a service using telecom dataset.
- Implemented Logistic Regression, Decision Tree and Random Forest algorithms.
- Performed data cleaning, feature engineering and model evaluation using confusion matrix and ROC curve.
- Achieved more than 85% prediction accuracy using Random Forest classifier.

AI Chatbot using LLM — Python, NLP

- Developed an AI chatbot capable of answering user queries using Large Language Model APIs.
- Implemented prompt engineering techniques to improve response accuracy and contextual understanding.
- Integrated natural language processing techniques for question understanding and response generation.
- Designed a conversational interface capable of handling multiple user queries effectively.

Experience

Korba Hospital – Diagnostic Lab Intern

- Assisted in clinical diagnostic procedures including blood analysis, urinalysis, microbial culture and ELISA tests.
- Operated laboratory equipment such as centrifuge, microscope and biochemical analyzers.
- Maintained patient diagnostic records and supported laboratory data management.

Achievements

- Developed interdisciplinary projects integrating Biotechnology with Machine Learning applications.
- Hands-on experience with data analysis, machine learning models and AI based problem solving.
- Built multiple end-to-end ML projects including recommender systems, predictive models and AI chatbots.

Languages

English, Hindi